

Differential Equations Lesson 05

Question 1:

$$f(x, y) = 2x - 15y + 8$$

$$\text{a) } \frac{\partial f(x, y)}{\partial x} = f_x(x, y) = \underline{\hspace{2cm} ? \hspace{2cm}}$$

$$\text{b) } \frac{\partial f(x, y)}{\partial y} = f_y(x, y) = \underline{\hspace{2cm} ? \hspace{2cm}}$$

Question 2:

$$f(x, y) = 2x^4 y^3 - 5y$$

$$\text{a) } \frac{\partial f(x, y)}{\partial x} = f_x(x, y) = \underline{\hspace{2cm} ? \hspace{2cm}}$$

$$\text{b) } \frac{\partial f(x, y)}{\partial y} = f_y(x, y) = \underline{\hspace{2cm} ? \hspace{2cm}}$$

Question 3:

$$f(x, y) = x^2 \sqrt{y}$$

$$\text{a) } \frac{\partial f(x, y)}{\partial x} = f_x(x, y) = \underline{\hspace{2cm} ? \hspace{2cm}}$$

$$\text{b) } \frac{\partial f(x, y)}{\partial y} = f_y(x, y) = \underline{\hspace{2cm} ? \hspace{2cm}}$$

Question 4:

$$f(x, y) = 2x^2 e^{xy}$$

$$\text{a) } \frac{\partial f(x, y)}{\partial x} = f_x(x, y) = \underline{\hspace{2cm} ? \hspace{2cm}}$$

$$\text{b) } \frac{\partial f(x, y)}{\partial y} = f_y(x, y) = \underline{\hspace{2cm} ? \hspace{2cm}}$$

Question 5:

$$f(x, y) = \ln(2x^2 - 5y + 3)$$

$$\text{a) } \frac{\partial f(x, y)}{\partial x} = f_x(x, y) = \underline{\hspace{2cm} ? \hspace{2cm}}$$

$$\text{b) } \frac{\partial f(x, y)}{\partial y} = f_y(x, y) = \underline{\hspace{2cm} ? \hspace{2cm}}$$

Question 6:

$$f(x, y) = \cos(3x + 5y)$$

$$\text{a) } \frac{\partial f(x, y)}{\partial x} = f_x(x, y) = \underline{\hspace{2cm} ? \hspace{2cm}}$$

$$\text{b) } \frac{\partial f(x, y)}{\partial y} = f_y(x, y) = \underline{\hspace{2cm} ? \hspace{2cm}}$$

Question 7:

$$f(x, y) = 2x^2y + C(y); \quad \text{where } C(y) = 4y^2 + 7$$

$$\text{a) } \frac{\partial f(x, y)}{\partial x} = f_x(x, y) = \underline{\hspace{2cm}} ?$$

$$\text{b) } \int f_x(x, y) \cdot dx = \underline{\hspace{2cm}} ?$$

Question 8:

$$f(x, y) = 4 \ln x + C(y); \quad \text{where } C(y) = y^3 + 3y + 2$$

$$\text{a) } \frac{\partial f(x, y)}{\partial x} = f_x(x, y) = \underline{\hspace{2cm}} ?$$

$$\text{b) } \int f_x(x, y) \cdot dx = \underline{\hspace{2cm}} ?$$

Question 9:

$$f(x, y) = \sin(5x + 2y)$$

a) $\frac{\partial f(x, y)}{\partial x} = f_x(x, y) = \underline{\hspace{2cm}} ?$

b) $\int f_x(x, y) \cdot dx = \underline{\hspace{2cm}} ?$

Question 10:

$$f(x, y) = 2x^5 y + D(x); \quad \text{where } D(x) = 4x^2 + 1$$

$$\text{a) } \frac{\partial f(x, y)}{\partial y} = f_y(x, y) = \underline{\hspace{2cm}} ?$$

$$\text{b) } \int f_y(x, y) \cdot dx = \underline{\hspace{2cm}} ?$$

Question 11:

$$f(x, y) = \cos(x^2 + y^2) + D(x);$$

$$\text{where } D(x) = x^3 + 1$$

$$\text{a) } \frac{\partial f(x, y)}{\partial y} = f_y(x, y) = \underline{\hspace{2cm}}?$$

$$\text{b) } \int f_y(x, y) \cdot dx = \underline{\hspace{2cm}}?$$

Question 12:

$$F(x, y(x)) = 3x^3y + 3y \quad \text{and} \quad y(x) = 5x^2$$

Find $D_x[F(x, y(x))]$.

Hint: Chain Rule: $D_x[F(x, y(x))] = \frac{dF(x, y(x))}{\partial x} + \frac{dF(x, y(x))}{\partial y} \cdot \frac{dy}{dx}$

a) $\frac{dF(x, y(x))}{\partial x} = \underline{\hspace{2cm}} ?$

b) $\frac{dF(x, y(x))}{\partial y} = \underline{\hspace{2cm}} ?$

c) $D_x[F(x, y(x))] = \frac{dF(x, y(x))}{\partial x} + \frac{dF(x, y(x))}{\partial y} \cdot \frac{dy}{dx} = \underline{\hspace{2cm}} ?$

Question 13:

Hint: Chain Rule: $D_x[F(x, y(x))] = \frac{dF(x, y(x))}{\partial x} + \frac{dF(x, y(x))}{\partial y} \cdot \frac{dy}{dx}$

a) $\frac{dF(x, y(x))}{\partial x} = \underline{\hspace{2cm}} ?$

b) $\frac{dF(x, y(x))}{\partial y} = \underline{\hspace{2cm}} ?$

c) $D_x[F(x, y(x))] = \frac{dF(x, y(x))}{\partial x} + \frac{dF(x, y(x))}{\partial y} \cdot \frac{dy}{dx} = \underline{\hspace{2cm}} ?$

Question 14:

$$F(x, y(x)) = x^3 + 3y^4 \quad \text{and} \quad y(x) = \tan x$$

Find $D_x[F(x, y(x))]$.

Hint: Chain Rule: $D_x[F(x, y(x))] = \frac{dF(x, y(x))}{\partial x} + \frac{dF(x, y(x))}{\partial y} \cdot \frac{dy}{dx}$

a) $\frac{dF(x, y(x))}{\partial x} = \underline{\hspace{2cm}}?$

b) $\frac{dF(x, y(x))}{\partial y} = \underline{\hspace{2cm}}?$

c) $D_x[F(x, y(x))] = \frac{dF(x, y(x))}{\partial x} + \frac{dF(x, y(x))}{\partial y} \cdot \frac{dy}{dx} = \underline{\hspace{2cm}}?$

Question 15

Let $w = x^4 \cdot y - 6y$, where $x(t) = \sin t$ and $y(t) = e^t$.

Find $\frac{dw}{dt}$.

Using Direct Method:

a) Write w as function of t only: _____?

b) $\frac{dw}{dt} =$ _____?

Using Chain Rule: $\frac{dw}{dt} = \frac{\partial w}{\partial x} \cdot \frac{dx}{dt} + \frac{\partial w}{\partial y} \cdot \frac{dy}{dt}$

c) $\frac{\partial w}{\partial x} =$ _____?; $\frac{dx}{dt} =$ _____?; $\frac{\partial w}{\partial y} =$ _____?; $\frac{dy}{dt} =$ _____?

d) $\frac{dw}{dt} = \frac{\partial w}{\partial x} \cdot \frac{dx}{dt} + \frac{\partial w}{\partial y} \cdot \frac{dy}{dt} =$ _____?

